

SADIA BINTE MONIR

 github.com/monir405

 sbmonir@mun.ca

 linkedin.com/in/sadia-binte-monir

WHY CHOOSE ME

- 3+ years of software development experience building backend, full-stack, and data-driven applications, with strong project experience in Python, TypeScript, REST APIs, PostgreSQL, Redis, and Docker.
- Hands-on experience building cloud-integrated projects using AWS services including S3, Lambda, EMR Serverless, Glue Data Catalog, Athena, Lake Formation, and IAM, with infrastructure provisioned through Terraform..
- AWS Certified Cloud Practitioner with knowledge of cloud architecture, IAM, VPCs, subnets, security groups, and core networking/security concepts

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** (February 8, 2026)
- **Data Analysis With Python** by freeCodeCamp (January 20, 2024).
- **Responsive Web Design** by freeCodeCamp (October 28, 2023).
- **Scientific Computing With Python** by freeCodeCamp (October 12, 2023).

TECHNOLOGIES

- Languages: Python, TypeScript, JavaScript, Java, C/C++, React, Next.js, React Native, HTML5, CSS3, Material UI
- Backend & APIs: FastAPI, NestJS, Express, REST APIs
- Data & Databases: PostgreSQL, MySQL, Redis, Pandas
- Cloud & Infrastructure: AWS (S3, Lambda, IAM, VPC, Glue, Athena, EMR Serverless, Lake Formation), Terraform, Docker
- Tools & Platforms: Git, GitHub, Linux, Windows

EDUCATION

Software Development Co-op
College Of North Atlantic
09/2023 - 04/2026,

EXPERIENCES

Corsphere | Software Developer 2022 - Present

Sensor Data Pipeline: Developed a configurable data pipeline system for cleaning, validating, and aligning time-series sensor data, focusing on data quality, consistency, and scalable processing workflows.

Technologies Used: Python, FastAPI, PostgreSQL, Pandas, Poetry, YAML, Terraform, AWS

- Built a configurable sensor-data ETL platform in Python and FastAPI to ingest CSV telemetry, enforce data-quality rules, and align time-series datasets for downstream analytics.
- Implemented an **S3**-backed raw, cleaned, and curated data lake, persisting processed outputs and run artifacts to support repeatable processing and historical analysis.
- Added event-driven ingestion with **S3**, **Lambda**, and **boto3**-based orchestration, and used **PySpark** on **EMR** Serverless for batch transformations and backfill workflows
- Integrated **AWS Glue Data Catalog**, **Athena**, and **Lake Formation** to publish curated datasets for SQL-based analysis with governed access controls.
- Provisioned the platform with **Terraform**, including **IAM** roles and policies, **VPC** networking, subnets, security groups, and deployment of a Dockerized FastAPI control-plane service.

P&ID (Piping and Instrumentation Diagram) Visualization: Developed an interactive P&ID visualization system, enabling users to design, manipulate, and export diagrams. Focused on real-time updates and persistent state storage for diagram edits.

Technologies Used: React, React Flow, Next.js, PostgreSQL, Jotai, TypeScript

- Built an interactive P&ID editor for users to create, update, and persist diagram state with immediate UI feedback.
- Designed and implemented backend APIs for diagram save and load workflows, storing structured diagram data in **PostgreSQL** for reliable retrieval and editing.
- Implemented diagram export handling with **AWS S3**, including secure upload and retrieval flows using presigned URLs and **IAM-based** access control.
- Added an **AWS Lambda** workflow to process exported diagram artifacts asynchronously, supporting validation and metadata generation for JSON/PNG outputs.
- Provisioned core cloud resources with Terraform, defining reproducible infrastructure for **S3**, **Lambda**, and **IAM** policies

Mapper - Voyage Visualization & Planning System: Developed an advanced maritime system to plan, track, and optimize voyages with interactive maps, dynamic route visualization, and real-time weather integration.

Technologies Used: React, React-Leaflet, Mapshaper, PostgreSQL, Redis, Next.js, Weather APIs, GeoJSON, Supercluster

- Built an S3-backed geospatial asset pipeline for voyage planning data, storing uploaded GeoJSON and CSV inputs and publishing processed map layers for frontend consumption.
- Implemented a serverless preprocessing workflow with **AWS Lambda** and **Mapshaper** to normalize and optimize route overlays, reducing client-side map payloads and improving layer load performance.
- Automated weather and **NOTMAR** data refresh using Amazon **EventBridge Scheduler**, updating voyage overlays on a recurring schedule for near-real-time planning visibility.
- Deployed the Next.js and React-Leaflet application on **AWS Amplify**, supporting interactive waypoint routing, clustered map layers, and timeline-based voyage visualization.
- Provisioned core AWS infrastructure with **Terraform**, including S3, Lambda, IAM roles/policies, and scheduled jobs for repeatable deployment of the mapping workflow.

Additional Projects:

- **Team Performance Dashboard:** Built a full-stack dashboard for engineering managers to monitor sprint metrics, team performance, and morale, with real-time **Redis/WebSocket** updates, CSV ingestion, PDF export, and lightweight **AWS S3/Terraform** support.
- **Graph Visualizer:** Developed a graphing tool for uploading structured CSVs, normalizing data, and visualizing trends over time; engineered chunked median sampling and dynamic axis scaling to improve performance on large datasets.
- **Scanner UI & Real-Time Notification Feed:** Built a scanner UI with multi-camera support, autofocus, detection overlays, and Redis-backed real-time notifications, with state-driven controls for smooth event-based interactions.